

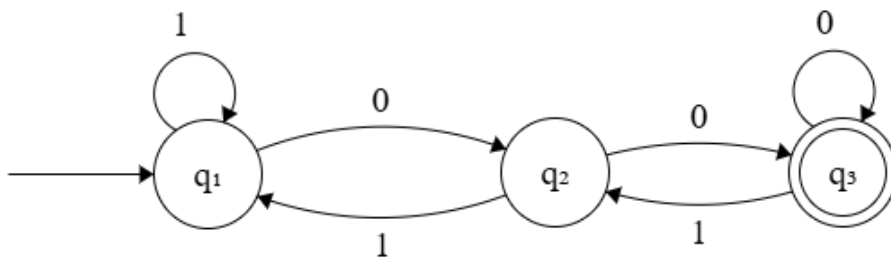
Oefeningen talen en automaten - Reeks 5-6

Oefeningenopdracht 3: opgave en indienen

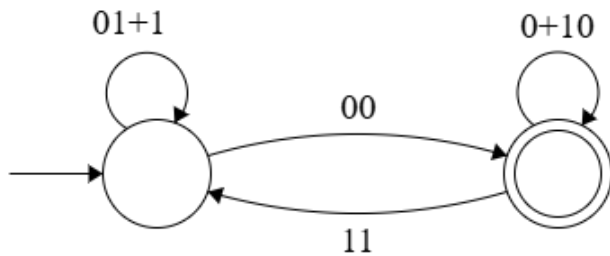
Reeks 5

1.

DFA



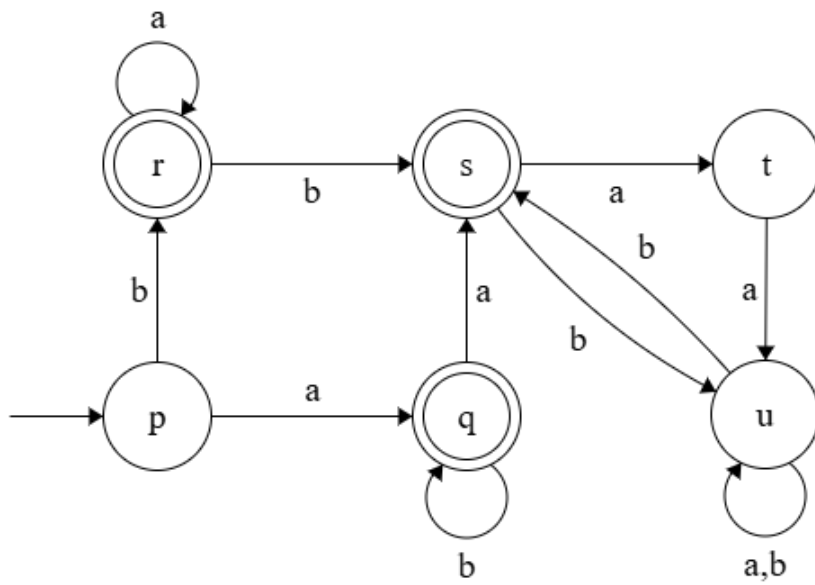
REGEX



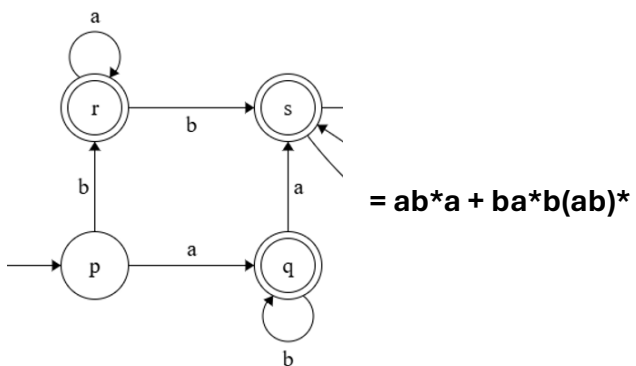
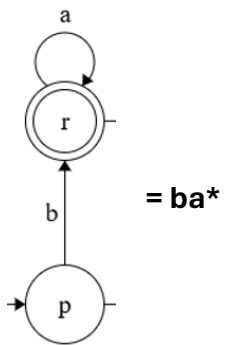
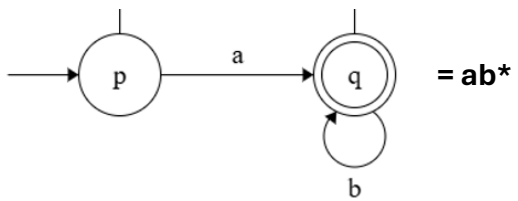
$((1 + 01) + 00 (0 + 10)^* 11)^* 00(0+10)^*$

2.

DFA



REGEX



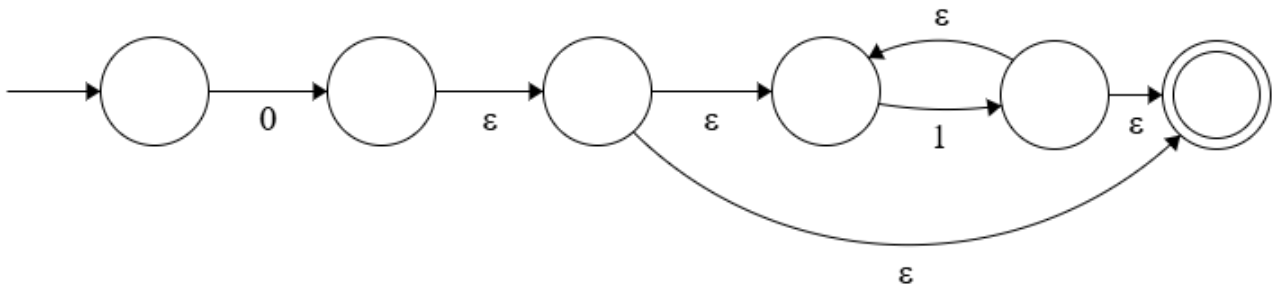
→ $Ab^* + ba^* + (ab^*a + ba^*b(ab)^*)$

3.

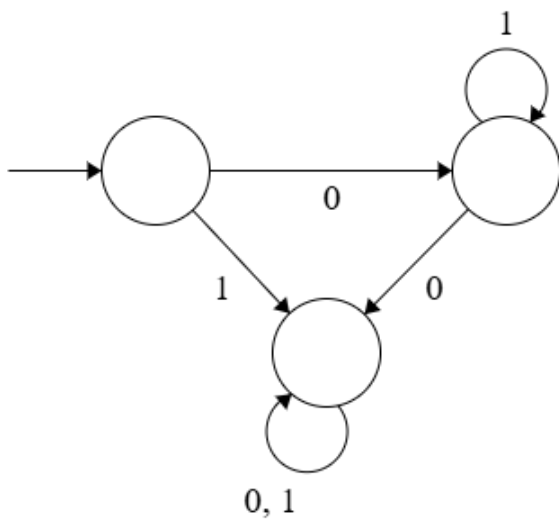
$L(A) = X \rightarrow L(A_{e\text{-NFA}}) = XX^*$

4.

e-NFA



DFA



5.

Reeks 6

1.

TB: $L = \{0^r 10^r \mid r \geq 1\}$ is niet regulier

Bewijs:

Stel L regulier $\Rightarrow \exists n$ pumpings-lemma

Neem $w = 0^n 1 0^n \in L$, $|w| > n$

$\Rightarrow$ we schrijven w als xyz met:

$y \neq \varepsilon$

$|xy| \leq n$

$\forall i \geq 0 \ x y^i z \in L$ ($\Rightarrow x$ en y bevatten enkel 0'en)

$\Rightarrow$ stel $i=0$

$\Rightarrow x y^0 z = x z \in L$

$0^m 1 0^n \in L$ met $m < n$

$\Rightarrow y$ is weg, dus minder a'en

$\Rightarrow$ contradictie : ieder element van taal moet $|x|$ en $|y|$ zonder gelijken hebben zijn. Dus de taal is niet regulier.